

# Control Charts

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## Introduction to Control Charts

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A **Control Chart** is a specialized line graph used in **Statistical Quality Control (SQC)** to monitor a process over time. Its main purpose is to distinguish between two types of process variation:

1. **Common Cause Variation (or Random Variation):** The natural, expected, inherent variability in any stable process. This variation is small and random, and trying to eliminate it is usually costly and ineffective.
2. **Special Cause Variation (or Assignable Cause Variation):** Unexpected, non-random shifts or changes in the process due to specific identifiable factors (e.g., a broken machine part, a new operator, a bad batch of raw material). This type of variation indicates the process is **"out of control"** and requires investigation and correction.

The process variable being measured (e.g., the average weight of a product, the number of defects) is plotted on the chart over time .

The objective of using a control chart is twofold:

- To find any **"special" causes of variation** so they can be identified and eliminated, bringing the process back into a state of statistical control.
- To reflect the **process improvements** that have been made, showing reduced variability or a shift toward the target value.



## Statistical Quality Control (SQC)

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**Statistical Quality Control (SQC)** is the application of statistical methods and techniques to the monitoring and controlling of quality in the manufacturing and service industries. It ensures that products and services meet the required specifications and standards.

## Methods of SQC

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SQC methods are generally categorized into two areas:

1. **Process Control:** This is concerned with **preventing defects** by monitoring the process while it is running. The primary tool for Process Control is the **Control Chart**.
  - **Goal:** To maintain the process in a state of statistical control so that only common cause variation is present.
2. **Product Control (or Acceptance Sampling):** This is concerned with **checking finished goods** to decide whether a batch (or lot) of products should be accepted or rejected based on inspecting a small sample from that batch.
  - **Goal:** To determine the quality of a batch of goods before or after a transaction.



## Control Chart Terminology

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Every standard control chart features three horizontal lines that define the expected range of the process.

1. **Center Line (CL):**
  - **Definition:** Represents the **average** or **target value** of the quality characteristic being measured (e.g., the true population mean,  $\mu$ , or the overall grand mean of the process data,  $\bar{\bar{x}}$ ).
  - **Purpose:** It serves as the baseline for the process.

## 2. Upper Control Limit (UCL):

- **Definition:** The line drawn **above** the Center Line, typically set at **three standard deviations** ( $+3\sigma$ ) above the mean.
- **Purpose:** Points falling above the UCL signal that a **special cause of variation** is likely affecting the process, causing the characteristic to be unusually high.

## 3. Lower Control Limit (LCL):

- **Definition:** The line drawn **below** the Center Line, typically set at **three standard deviations** ( $-3\sigma$ ) below the mean.
- **Purpose:** Points falling below the LCL signal that a **special cause of variation** is likely affecting the process, causing the characteristic to be unusually low.

// **Note on 3-Sigma Limits:** The  $\pm 3\sigma$  limits mean that if the process is normally distributed and only common causes are present, approximately **99.73%** of all data points should fall between the UCL and LCL. Any point outside these limits is extremely unlikely to be random chance and is taken as evidence of a special cause.

# Mean and Range Control Charts ( $\bar{x}$ and $R$ Charts)

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Control charts are categorized based on the type of data they monitor. The **Mean ( $\bar{x}$ ) and Range ( $R$ ) charts** are used together for **variable data**, which is continuous and measurable (e.g., weight, height, temperature). They are applied to data collected in small, fixed-size subgroups over time.

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## Mean Chart ( $\bar{x}$ Chart)

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- **Purpose:** To monitor the **central tendency** (the average) of the process. It tells you if the **average level** of the process is shifting up or down.
- **Central Line ( $CL_{\bar{x}}$ ):** Represents the overall process average ( $\bar{\bar{x}}$ , or  $\mu$  if known).
- **Control Limits ( $UCL_{\bar{x}}$  and  $LCL_{\bar{x}}$ ):** Define the expected range for the subgroup means.

The control limits are calculated using the average range ( $\bar{R}$ ):

$$UCL_{\bar{x}} = \bar{\bar{x}} + A_2 \bar{R}$$

$$LCL_{\bar{x}} = \bar{\bar{x}} - A_2 \bar{R}$$

- $A_2$  is a constant determined by the subgroup size ( $n$ ).

## Range Chart ( $R$ Chart)

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- **Purpose:** To monitor the **process variability** (the spread or consistency) within the subgroups. It tells you if the **consistency** of the process is increasing or decreasing.
- **Central Line ( $CL_R$ ):** Represents the average range ( $\bar{R}$ ) of the subgroups.
- **Control Limits ( $UCL_R$  and  $LCL_R$ ):** Define the expected range for the subgroup ranges.

The control limits are calculated using constants  $D_4$  and  $D_3$ :

$$UCL_R = D_4 \bar{R}$$

$$LCL_R = D_3 \bar{R}$$

- $D_4$  and  $D_3$  are constants determined by the subgroup size ( $n$ ).

**// Important:** The Range chart **must be analyzed first**. If the process variation is out of control (the  $R$  chart is out of control), the limits calculated for the  $\bar{x}$  chart are not valid.

# $\bar{x}$ and $R$ Chart Examples

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We will assume a common table of control chart constants:

| Subgroup Size ( $n$ ) | $A_2$ | $D_4$ | $D_3$ |
|-----------------------|-------|-------|-------|
| 4                     | 0.729 | 2.282 | 0     |
| 5                     | 0.577 | 2.114 | 0     |

## Example 1: Calculating Control Limits ( $n = 4$ )

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A process is monitored using 10 subgroups of size  $n = 4$ . The total of the subgroup means ( $\sum \bar{x}$ ) is 420, and the total of the subgroup ranges ( $\sum R$ ) is 30. Calculate the control limits for the  $\bar{x}$  and  $R$  charts.

### 1. Calculate Averages ( $\bar{\bar{x}}$ and $\bar{R}$ )

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- Number of subgroups ( $k$ ): 10
- Overall Mean ( $\bar{\bar{x}}$ ):**

$$\bar{\bar{x}} = \frac{\sum \bar{x}}{k} = \frac{420}{10} = \mathbf{42.0}$$

- Average Range ( $\bar{R}$ ):**

$$\bar{R} = \frac{\sum R}{k} = \frac{30}{10} = \mathbf{3.0}$$

### 2. Determine Constants

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For subgroup size  $n = 4$ :  $A_2 = 0.729$ ,  $D_4 = 2.282$ ,  $D_3 = 0$ .

### 3. Calculate $R$ Chart Control Limits

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- $CL_R = \bar{R} = \mathbf{3.0}$
- $UCL_R = D_4 \bar{R} = 2.282 \times 3.0 = \mathbf{6.846}$
- $LCL_R = D_3 \bar{R} = 0 \times 3.0 = \mathbf{0}$

### 4. Calculate $\bar{x}$ Chart Control Limits

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- $CL_{\bar{x}} = \bar{\bar{x}} = \mathbf{42.0}$
- $UCL_{\bar{x}} = \bar{\bar{x}} + A_2 \bar{R} = 42.0 + (0.729 \times 3.0) = 42.0 + 2.187 = \mathbf{44.187}$
- $LCL_{\bar{x}} = \bar{\bar{x}} - A_2 \bar{R} = 42.0 - (0.729 \times 3.0) = 42.0 - 2.187 = \mathbf{39.813}$



## Example 2: Analyzing Data and Interpreting Results ( $n = 5$ )

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Five samples (subgroups) of size  $n = 5$  were taken from a production process. The subgroup means ( $\bar{x}$ ) and ranges ( $R$ ) are given below. Determine the control limits and comment on the process status. (Use  $n = 5$  constants:  $A_2 = 0.577, D_4 = 2.114, D_3 = 0$ ).

| Subgroup  | 1    | 2    | 3   | 4    | 5    | $\Sigma$ |
|-----------|------|------|-----|------|------|----------|
| $\bar{x}$ | 10.2 | 11.0 | 9.8 | 10.4 | 12.1 | 53.5     |
| $R$       | 1.8  | 2.5  | 1.9 | 2.1  | 4.0  | 12.3     |

### 1. Calculate Averages ( $\bar{\bar{x}}$ and $\bar{R}$ )

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- Number of subgroups ( $k$ ): 5
- **Overall Mean ( $\bar{\bar{x}}$ ):**

$$\bar{\bar{x}} = \frac{53.5}{5} = 10.7$$

- **Average Range ( $\bar{R}$ ):**

$$\bar{R} = \frac{12.3}{5} = 2.46$$

## 2. Calculate Control Limits

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### A. $R$ Chart Limits:

- $CL_R = 2.46$
- $UCL_R = D_4\bar{R} = 2.114 \times 2.46 \approx 5.20$
- $LCL_R = D_3\bar{R} = 0 \times 2.46 = 0$

### B. $\bar{x}$ Chart Limits:

- $CL_{\bar{x}} = 10.7$
- $UCL_{\bar{x}} = \bar{\bar{x}} + A_2\bar{R} = 10.7 + (0.577 \times 2.46) = 10.7 + 1.42 = 12.12$
- $LCL_{\bar{x}} = \bar{\bar{x}} - A_2\bar{R} = 10.7 - 1.42 = 9.28$

## 3. Analyze and Interpret

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### A. $R$ Chart Analysis (Variability):

- Compare the subgroup ranges ( $R$ ) to the  $R$  chart limits (0 to 5.20).
- All ranges (1.8, 2.5, 1.9, 2.1, 4.0) are **within** the limits.
- **Conclusion for  $R$  Chart:** The process variability (spread) is **in statistical control**.

### B. $\bar{x}$ Chart Analysis (Mean):

- Compare the subgroup means ( $\bar{x}$ ) to the  $\bar{x}$  chart limits (9.28 to 12.12).
- Subgroup 5 has a mean of **12.1**, which is very close but **below**  $UCL_{\bar{x}} = 12.12$ .
- **Conclusion for  $\bar{x}$  Chart:** All subgroup means are **within** the limits.

**Overall Process Status:** Both the  $\bar{x}$  chart and the  $R$  chart are in control. The process is operating in a state of **statistical control**.

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## Control Charts for Attributes

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**Attribute charts** are used for data that are **discrete** and result from **counting** rather than measurement. This data typically falls into two categories:

1. **Fraction Defective or Proportion Nonconforming ( $p$  and  $np$  Charts):**  
Counting the proportion of **items** that are defective (i.e., items that fail to meet specifications).
2. **Count of Defects or Nonconformities ( $c$  and  $u$  Charts):** Counting the number of **defects** per item (e.g., scratches on a car, errors in a page of text).

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### 1. $p$ -Chart (Fraction Defective)

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The  **$p$ -chart** monitors the **proportion or fraction of nonconforming (defective) items** in a series of subgroups, typically when the subgroup size ( $n$ ) **varies**.

### How to Solve the $p$ -Chart

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1. **Calculate Overall Proportion ( $\bar{p}$ ):** This is the Central Line (CL).

$$\bar{p} = \frac{\text{Total number of defective items}}{\text{Total number of items inspected}}$$

2. **Calculate Control Limits:** The limits are based on the standard deviation of the binomial distribution, estimated using  $\bar{p}$ .

$$CL = \bar{p}$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_j}}$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_j}}$$

*If  $n_j$  (subgroup size) varies, the limits must be calculated separately for each subgroup.*

## **$p$ -Chart Examples**

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### **Example 1: $p$ -Chart with Constant Subgroup Size**

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A factory inspects 10 subgroups of 100 parts each ( $n = 100$ ). The total number of defective parts found across all subgroups is 20. Determine the control limits.

1. **Calculate  $\bar{p}$  (CL):**

- Total inspected:  $10 \times 100 = 1000$
- Total defective: 20

$$\bar{p} = \frac{20}{1000} = \mathbf{0.02}$$

$$CL = 0.02$$

2. **Calculate Control Limits ( $n = 100$ ):**

$$UCL = 0.02 + 3\sqrt{\frac{0.02(1 - 0.02)}{100}} = 0.02 + 3\sqrt{\frac{0.0196}{100}}$$

$$UCL = 0.02 + 3\sqrt{0.000196} = 0.02 + 3(0.014) = 0.02 + 0.042 = \mathbf{0.062}$$

$$LCL = 0.02 - 0.042 = \mathbf{-0.022}$$

3. **Final Limits:** Since proportion cannot be negative,  $LCL = 0$ .

## Example 2: $p$ -Chart Interpretation

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In the above example, if Subgroup 5 had 8 defective parts. Is the process in control?

- Subgroup 5 proportion ( $p_5$ ):  $p_5 = 8/100 = 0.08$
- $UCL = 0.062$
- **Conclusion:** Since  $p_5(0.08) > UCL(0.062)$ , the process is **out of control**. A special cause of variation likely occurred during Subgroup 5's production.

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## 2. $np$ -Chart (Number of Defectives)

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The  $np$ -chart monitors the **actual number of nonconforming (defective) items** in a subgroup. It is only appropriate when the subgroup size ( $n$ ) is **constant**.

### How to Solve the $np$ -Chart

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1. **Calculate Average Number of Defectives ( $np$ ):** This is the Central Line (CL).

$$\overline{np} = \frac{\text{Total number of defective items}}{\text{Number of subgroups inspected}}$$

2. **Calculate Control Limits:** The limits use the average number of defectives directly.

$$CL = \overline{np}$$

$$UCL = \overline{np} + 3\sqrt{\overline{np}(1 - \bar{p})}$$

$$LCL = \overline{np} - 3\sqrt{\overline{np}(1 - \bar{p})}$$

*Note:  $\bar{p}$  is calculated as  $\overline{np}/n$ .*

## *np*-Chart Examples

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### Example 1: Calculating Control Limits (*n* constant)

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Eight subgroups, each of size  $n = 50$ , were inspected. The total number of nonconforming units found was 16. Calculate the control limits.

1. **Calculate  $np$  and  $\bar{p}$ :**

- $n = 50$ ,  $k = 8$ . Total inspected:  $8 \times 50 = 400$ .
- $\overline{np} = \frac{16}{8} = \mathbf{2.0}$  (Average number of defectives per subgroup).  $CL = 2.0$ .
- $\bar{p} = \frac{\text{Total defectives}}{\text{Total inspected}} = \frac{16}{400} = 0.04$

2. **Calculate Control Limits:**

$$UCL = 2.0 + 3\sqrt{2.0(1 - 0.04)} = 2.0 + 3\sqrt{2.0 \times 0.96}$$

$$UCL = 2.0 + 3\sqrt{1.92} = 2.0 + 3(1.386) = 2.0 + 4.158 = \mathbf{6.158}$$

$$LCL = 2.0 - 4.158 = \mathbf{-2.158}$$

3. **Final Limits:** Since count cannot be negative,  $LCL = 0$ .

### Example 2: *np*-Chart Interpretation

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If Subgroup 6 had **8 nonconforming units**. Is the process in control?

- Subgroup 6 count: 8
- $UCL = 6.158$
- **Conclusion:** Since the count (8)  $>$   $UCL(6.158)$ , the process is **out of control**.

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## 3. *c*-Chart (Count of Defects)

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The **c-chart** monitors the **number of defects (nonconformities)** found in a unit, sample, or area, where the unit size (area of opportunity) is **constant**. This assumes the defects follow a **Poisson distribution**.

## How to Solve the c-Chart

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1. **Calculate Average Number of Defects ( $\bar{c}$ ):** This is the Central Line (CL).

$$\bar{c} = \frac{\text{Total number of defects}}{\text{Total number of units inspected}}$$

2. **Calculate Control Limits:** The limits are based on the Poisson distribution, where the standard deviation is  $\sqrt{\bar{c}}$ .

$$CL = \bar{c}$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}}$$

## c-Chart Examples

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### Example 1: Calculating Control Limits (Constant Area)

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Twenty sample units (e.g., bolts of cloth) were inspected. The total number of defects found was 80. Calculate the control limits.

1. **Calculate  $\bar{c}$  (CL):**

- Total units inspected: 20
- Total defects: 80

$$\bar{c} = \frac{80}{20} = 4.0$$

$$CL = 4.0$$

2. **Calculate Control Limits:**

$$UCL = 4.0 + 3\sqrt{4.0} = 4.0 + 3(2) = 4.0 + 6 = \mathbf{10.0}$$

$$LCL = 4.0 - 3\sqrt{4.0} = 4.0 - 6 = \mathbf{-2.0}$$

3. **Final Limits:** Since count cannot be negative,  $LCL = 0$ .

## Example 2: $c$ -Chart Interpretation

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If Unit 15 has **11 defects**. Is the process in control?

- Unit 15 count ( $c_{15}$ ): **11**
- $UCL = 10.0$
- **Conclusion:** Since  $c_{15}(11) > UCL(10.0)$ , the process is **out of control**.